Expectation: Let $f(x)$ be the PMF of any distribut^n, then expectat^n of random variable x is defined as

$$\boxed{E[X] = \sum_{x=-\infty}^{\infty} x \cdot f(x)} \quad \text{— (1)}$$

In case of binomial distribut^n

$$f(x) = {}^nC_x (1-p)^{n-x} p^x$$

$$x = 0, 1, \cdots, n$$

$$\boxed{E[X] = \sum_{x=0}^{n} x \cdot {}^nC_x (1-p)^{n-x} p^x} \quad \text{— (2)}$$

$$\boxed{E[X] = np} \quad \text{(3)} \rightarrow \text{expectation of binomial distribut}^n$$

This expectation also gives the mean of the distribution.

$$\text{Mean} = E[X]$$

Now, we will find variance of the distribut^n.

$$\text{Var}(x) = E[X^2] - (E[X])^2.$$

In case of binomial distribut^n $\text{Var}(x) = np(1-p)$.

$$E[x^m] = \sum_{x=-\infty}^{\infty} x^m \cdot f(x)$$

$$E[g(x)] = \sum_{x=-\infty}^{\infty} g(x) \cdot f(x)$$

Poisson distributⁿ:

$$E[x] = \lambda \quad , \quad Var[x] = \lambda$$

Geometric distributⁿ:

$$E[x] = \frac{1}{p} \quad , \quad Var(x) = \frac{(1-p)}{p^2}$$

Skewness:
$$\gamma_1 = \frac{\mu_3}{(\mu_2)^{3/2}}$$

$$\boxed{\mu_r = E[(X-\mu)^r] = \sum_{n=-\infty}^{\infty} (x-\mu)^r \cdot f(x)}$$

$$\mu_3 \; ?? \quad , \quad \mu_2 \; ?? \Rightarrow \gamma_1$$

(a) $\gamma_1 = 0$: symmetric

(b) $\gamma_1 > 0$: Right skewed

(c) $\gamma_1 < 0$: Left skewed

Kurtosis : $\gamma_2 = \dfrac{\mu_4}{\mu_2^2} - 3$

ⓐ If $\gamma_2 = 0$ mesokurtic
ⓑ If $\gamma_2 > 0$ leptokurtic
ⓒ If $\gamma_2 < 0$ platykurtic.

$$\mu_2 = \sum_{x=-\infty}^{\infty} (x - E[x])^2 f(x)$$

$$\mu_3 = \sum_{x=-\infty}^{\infty} (x - E[x])^3 f(x)$$

$$\mu_4 = \sum_{x=-\infty}^{\infty} (x - E[x])^4 f(x)$$